

mcidoc Student Documentation

Daniel McGuiness

Table of Contents

1	Introduction	1
1.1	Purpose of Making	1
1.2	Currently Supported Formats.....	2
2	Configuring the Document.....	3
2.1	Notational Information	3
2.2	Global Options	3
2.3	Thesis Specific Options	4
2.3.1	Supervisor.....	4
2.3.2	Reviewer	4
2.3.3	Table of Contents.....	5
2.3.4	Example: Double Degree Programme	5
2.3.5	Example: Company Based Thesis	6
2.3.6	Example: BSc Thesis in English	6
2.3.7	Example: MasterArbeit auf Deutsch	6
3	Preparing a Thesis.....	7
3.1	Writing the Abstract(s).....	7
4	Preparing a Lab Report	8
5	Preparing a Presentation.....	9

1 Introduction

`mcidoc` is a $\text{\LaTeX}$ document class designed to provide students studying at **MCI Mechanics** with all their documentation needs, whether it is to write a simple report or to write a their graduation thesis¹ whether it is a single or a double degree.

As it stands the following document types are supported and are implemented for both **english** and **german**.

- Thesis
- Report
- Paper

All these aforementioned sub-classes will be explained in the following sections.

1.1 Purpose of Making

There are three (3) reasons for the creation of this documentclass.

The first purpose is to standardise and streamline document creation which abides to the standards set by the MCI with regards to:

- Font-size, figure captions, bibliography, . . .
- Margin spacing, header and footer spacing, text information, . . .
- Spacing between lines, linebreak settings, spacing of headers, . . .
- Language determined grammatical rules (when to hyphenate and caption text and proper quotation marks)
- Simple error-correction codes to minimise wrong entering of values,

and so on so forth.

Its second purpose is to create a repository to house a **standardised** class to minimise any friction between the work of the student and the representation of the content.

Care was given to how the class behaves to minimise its interaction with other packages the end-user might need for their work.

The document standard was also designed with specific customisation in hand by either configuring its settings through the options presented in the `\documentclass` and using the macro `\SetMCI{}` which its features are documented here.

The tertiary purpose is perhaps more in line of streamlining. While $\text{\LaTeX}$ is an excellent typesetting language with is very powerful² capabilities, it is nevertheless a language with its quirks and customs that may not prove to be very useful outside of its own domain and therefore the user should not be worried of these quirks.

This duty is mostly taken care of the class designer.

`\SetMCI key=val` [Function]
 Allows global configuration of the document. Takes various **key=val** options based on the documentclass state and other sub configurations.

¹ This is supported for both M.Sc and B.Sc degrees.

² $\text{\LaTeX}$ is a language known to be accidentally Turing complete (https://beza1e1.tuxen.de/articles/accidentally_turing_complete.html), meaning with enough time, it is possible to do all calculations any computer language can do. Therefore it is within the realm of possibility for one student to conduct their entire software work on $\text{\LaTeX}$.

1.2 Currently Supported Formats

As mentioned previously the following **state**'s are supported

- **Thesis:** The primary way the thesis is written. Supports different options depending on the grade of the language requirements.
- **Report:** Used in creating reports for lectures and lab sessions. Examples include writing a homework submission, or writing a lab report.

All the aforementioned documents are bundled up as a class called `mcidoc`.³

Important

This class has been tested on different computers running Windows and macOS with the compile being Lua $\text{\TeX}$. However, as with everything, there will always be edge cases and I am sure to that end, if you encounter any error please create an issue in Github Repo (<https://github.com/dTmC0945/C-MCI-LaTeX-Class-mcidoc/tree/main>).

Warning

While I will try to help with any problem this class might have, it is by no means I am providing any kind of WARRANTY or TECHNICAL SUPPORT. There will be documentation and some example documents which should compile without any additional configuration. Nevertheless, I can only help with the errors from code and not with your personal $\text{\LaTeX}$ development environment (i.e., $\text{\TeX}$ Studio).

state [Variable]

A configuration option to change the overall behaviour of `mcidoc` class. This value can take different options based on which formats are supported, which are shown at the beginning of this section.

The behaviour of the documentclass can be changed by setting `state=<val>` of the optional argument of the `mcidoc` documentclass.

³ Of course.

2 Configuring the Document

The entire configuration is handled by a single command `\SetMCI` which the end user can change the behaviour by changing specific keys. The accessible options changes depending on the `state` the `mcidoc` is in whereas some options are global and are accessible across all classes.

2.1 Notational Information

Before options are described, it is worth explaining the slight notational quirks the documentclass has.

Majority of the options given by the `\SetMCI` are by default either as `set` or `unset`.

- If a feature is `set` by default, then the option will be executed as long as the user does not alter its value.
- Conversely, if a `\SetMCI` value is `unset`, then it will not show in the document or alter anything within the document as long as the user does not change its behaviour.

PROPERTYRESULT

<code>set</code>	Enable a feature
<code>unset</code>	Disable a feature

2.2 Global Options

The following options are available across different states:

T_EXnician's Note: These options are defined as `\Global@` prefix.

`/Degree` [Variable]
Sets the **Degree** of the document which can change some subtitles of the documentation.

Current supported values are: `BSc`, `MSc`.

`/Language` [Variable]
Sets the **Language** of the document which changes the typographical rules of the document and correctly sets the document titles and labels.

Current supported values are: `EN` (english), `DE` (german).

`/Deparment` [Variable]
Sets the **Deparment** of the document.

Current supported values are: `MECH` (Mechatronics).

`/StudentID` [Variable]
Sets the **Student Number** (i.e., Matrikelnummer) of the document.

As it is a FH, the number **must** begin with the digit 5.

2.3 Thesis Specific Options

The following options are used in the thesis mode only. Some options may have the same name as other states (such as the **Report** and **Thesis** both have their own version of **Title**) These options will only be described under the parent option in which they were invoked.

2.3.1 Supervisor

Supervisor is the primary person when the student is conducting their thesis. This person can be an internal (i.e., MCI Lecturer, Researcher) or could be someone from a company.

An example of this options is shown as below:

```
...
Supervisor      = {%
    Company = {Ministry of Sillyputty}, % <- The company of the supervisor
    Prefix  = {Dipl-Eng.},
    Name    = {Colin Devon},
    Postfix  = {M.Sc},
},%
...
```

/Thesis/Supervisor/Company [Variable]

Inserts the company name which the supervisor belongs to. This option should only be used if the supervisor is **outside** MCI as using this option will require the insertion of the Section 2.3.2 [Reviewer], page 4, option as well.

Default is **unset**.

/Thesis/Supervisor/Prefix [Variable]

Prints prefix honorifics for the supervisor, if there is any. For example titles such as Dr. or Dipl-Eng should be added here.

Default is **unset**.

/Thesis/Supervisor/Name [Variable]

Prints the name of the supervisor.

Default is **unset**.

/Thesis/Supervisor/Postfix [Variable]

Prints post honorifics of the supervisor. For example, titles such as Ph.D should be added here

Default is **unset**.

2.3.2 Reviewer

Reviewer is the examiner when the student is conducting their thesis with a company. The reviewer can be considered as a liason between the student and the MCI.

An example usage of this is shown below

```
...          % <- Previously the Supervisor was
...          % defined so we add the Reviewer
Reviewer     = {%
    Prefix    = {Rev.},
```

```

        Name      = {Ted Marshall},
        Postfix    = {Esq.},
    },%
    ...

/Thesis/Reviewer/Prefix [Variable]
    Prints prefix honorifics for the Reviewer, if there is any. For example titles such as
    Dr. or Dipl-Eng should be added here.
    Default is unset.

/Thesis/Reviewer/Name [Variable]
    Prints the name of the reviewer.
    Default is unset.

/Thesis/Reviewer/Name [Variable]
    Prints post honorifics of the reviewer. For example, titles such as Ph.D should be
    added here
    Default is unset.

```

2.3.3 Table of Contents

Allows the control of how the table of contents is printed.

```

/Thesis/ToC/Primary [Variable]
    Prints the main Table of Contents. Default is set and is the first table of content to
    be printed right after the abstract.
    Default is set.

/Thesis/ToC/Figures [Variable]
    Prints the List of Figures right after Table of Contents on a newpage.
    Default is set.
    If user has no figures within their document, this should changed to unset.

/Thesis/ToC/Tables [Variable]
    Prints the List of Tables right after Contents on a newpage if /Thesis/ToC/Figures is
    unset or right after /Thesis/ToC/Figures and on the same page if space if available.
    Default is set.
    If user has no table within their document, this should changed to unset.

```

2.3.4 Example: Double Degree Programme

The following setting is defined for the student **Karl Hendrix** doing a double degree with ASU (Appalachian State University). An example `\SetMCI` option would be as follows:

```

\SetMCI{%
    StudentName    = {%
        Name = {Karl Hendrix}
    },%
    StudentID      = {52414238},

```

```

Degree          = MSc,
Language        = EN,
Department      = MECH,
Thesis = { %
  Title          = {The Effects of Amplifier Gain on Guitar Pedal Distortion},
  StudyProgram   = {Smart Technologies},
  Supervisor     = {%                               % <- The MCI Supervisor
    Prefix = {Prof.},
    Name   = {David R. M. Jazz},
    Postfix = {},
  },%
  Double-Degree = {
    Institution   = {ASU},                               % <- The University code for ASU
    Student-Number = {6712345},
    Supervisor    = {%                               % <- Supervisor at ASU
      Prefix = {Prof.},
      Name   = {Colin Stevens},
      Postfix = {},
    },%
  },
  Embargo        = {31.07.2031},                          % <- An embargo request
  Bibliography   = {%
    Location      = {references.bib},                      % <- Bibliography resides in the root
  },
},
}

```

2.3.5 Example: Company Based Thesis

2.3.6 Example: BSc Thesis in English

2.3.7 Example: MasterArbeit auf Deutsch

3 Preparing a Thesis

3.1 Writing the Abstract(s)

Depending on the language requirements the thesis must either have:

- An abstract in English (EN only)
- An abstract in German and then in English (DE only)

For example the following environment generatex an abstract for the english section of the astract.

```
\begin{Abstract}{EN}
```

In recent years with the improvements in power electronics the use of high speed applications have increased. The industry standard Caged-Rotor Induction Motor (CRIM) is not suitable for high speed due of its mechanical durability and thermal properties. The most common rotor type for this application is a Solid Rotor Induction Motor (SRIM). This class of rotors are known to have high mechanical durability and good thermal properties.

This project focuses on comparing a CRIM with 3 types of SRIM. These rotors will be tested on a induction motor (IM) standard stator. The windings will be changed in order to keep the current passing in the stator for each rotor constant. The CRIM and smooth type SRIM will be testedn in ITÜ Elektrik Makinaları Laboratory and their high speed applications will be compared.

```
\end{Abstract}
```

4 Preparing a Lab Report

5 Preparing a Presentation